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December 2, 1974
SUPERSEDING
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December 7, 1971

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON (SINGLE AND PLIED FILLING YARNS, FLAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for unbleached flat duck.

1.2 Classification.

1.2.1 Types. The flat duck cotton cloth shall be of the following types as specified (see 6.2):

Type I	-	Single filling, 10.0 ounces
Type II	-	Double filling (plied), 7.3 ounces
Type III	-	Single filling, 8.0 ounces

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276	-	Thread, Cotton
PPP-P-1135	-	Packaging and Packing of Duck Fabrics (Cotton: Synthetic Fiber; Cotton-Synthetic Fiber Blends)

Federal Standards:

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitching

FSC 8305

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Yarns. Yarn shall be made from cotton which has been carded, drawn and spun.

3.2 Physical requirements. The cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in table I when tested as specified in 4.4.

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TABLE I. Physical requirements

Type	Weight per square yard minimum	Yarn ply		Yarns per inch minimum		Breaking strength 2/ pounds minimum	
	Ounces	Warp	Filling	Warp 1/	Filling	Warp	Filling
I	10.0	1	1	74	27	130	85
II	7.3	1	2	86	30	100	60
III	8.0	1	1	52	30	105	105

1/ Two yarns woven as one.

2/ A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

3.2.1 Weave. The weave shall be oxford, 2 ends weaving as one (plain weave).

3.3 Width. The width, inclusive of selvage, shall be as specified (see 6.2). A tolerance of minus 1/4 inch will be allowed on the specified width.

3.4 Nonfibrous materials. The starch and protein content, including chloroform-soluble and water-soluble material of the finished cloth shall be no more than 12.0 percent when tested as specified in 4.4.

3.5 Length and put-up. The cloth shall be put up in rolls in accordance with PPP-P-1135. Unless otherwise specified (see 6.2), the length of individual pieces shall be not less than 40 yards. Pieces shall be sewn together with a butted FSa-1 type seam of FED-STD-751, using ticket No. 10, 3 ply unbleached cotton thread conforming to type IIIA of V-T-276, and having 6 to 8 stitches per inch. The stitching shall be placed 1/2 ± 1/16 inch from the cut or raw edge of the cloth. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches of the roll on the outer surface.

3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.7 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.8 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.3.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section 1 of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.2, with the following exceptions: glossary numbers 1, 2, 3, 5, 14, 26, 27, and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of the warp yarns. Each knot or slub shall be scored as a defect only when it exceeds three times the normal yarn diameter. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 60 points. If one roll exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

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$$\frac{\text{Total points scored in sample size X 3600}}{\text{Contracted width of cloth (inches) X total yard inspected}} = \text{Points per 100 square yards}$$

4.2.2.2 Demerit points. Demerit Points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width - more than 1/4 inch less than specified.
Excessive neppiness.

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for put-up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present:

Defects

Any length found to be more than 2 yards less than the length marked on the roll ticket.

Length of piece:

(a) Any piece less than 40 yards in length.
(b) Pieces not sewn together with a butted FSa-1 seam.
(c) 6 to 8 stitches per inch not used in seam.
(d) Stitching not 1/2 ± 1/16 inch from the cut or raw edge of the cloth.

Number of pieces in roll not stenciled on last 6 inches of rolls on outer surface.

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4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths on the tickets.

4.2.3 Examination for compliance with Textile Fiber Products Identification Act and Face Marking. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.
Face stamping missing from either or both ends.
Face stamping on wrong side.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine whether packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1-1/2 yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Material identification	3.1.1	1200 1/
Yarn ply:		
Warp	3.2	Visual 2/
Filling	3.2	Visual 2/

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TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Weight	3.2	5041
Breaking strength:		
Warp	3.2	5100
Filling	3.2	5100
Yarns per inch:		
Warp	3.2	5050
Filling	3.2	5050
Weave	3.2.1	Visual 2/
Nonfibrous materials	3.4	2611

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1135.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for the following:

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Types I and III are intended for use as insole reinforcements of various footwear items.
Type II is intended for use in the following specifications:
UU-B-355 - Binder, Loose-Leaf, Prong
UU-B-356 - Binder, Loose-Leaf, Ring

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type of cloth required (see 1.2.1 and 3.2).
(c) Width of cloth required (see 3.3).
(d) Length, if other than specified (see 3.6).
(e) Selection of applicable levels of put-up packaging and packing (see 5.1 and 5.2).
(f) Whether special marking is required (see 5.3).

Custodians:

Army - GL
Navy - SA

Review activity:

Army - MD

User activity:

Navy - YD

Preparing activity:

Army - GL

CIVIL AGENCIES COORDINATING ACTIVITIES:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTEPPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

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